

C4: Yingchuan parameter checkpoint lambda=0.4, beta=0.0005

Sapelo array 48468709; five prespecified exploratory seeds. This is one of eight sequential checkpoints. All five jobs are terminal before this report; no next beta was launched for its construction.

Evidence boundary: measured CIFAR-100-C severity-5 trajectories and finite intervention responses. Numerical failure is a failed computation, not behavioral deterioration. A finite F response does not identify a closed Jacobian, spectrum or phase transition.

Complete five-seed status

Seed	Parent blocks	Extension blocks	Forks verified	H2O analyses	Coverage / reason
202	15	30	4/4	2/2	complete
303	15	30	4/4	2/2	complete
404	15	30	4/4	2/2	complete
505	15	30	4/4	2/2	complete
606	15	30	4/4	2/2	complete

A component is reported only after its own completion and verification. Parent, extension, fork and H2O gaps remain explicit; the raw run and failed attempts remain on Sapelo.

Exploratory readout: visit-3 minus visit-1 same-domain CE is positive in 4/5 seeds (mean +0.081); classification error changes by -3.79 percentage points on average. At H1410, 4/10 correlated parent nodes have harmful closed acceptance and 4/10 have positive feedback interaction. The signs are mixed; these are descriptive, not a persistent-deterioration or stability claim.

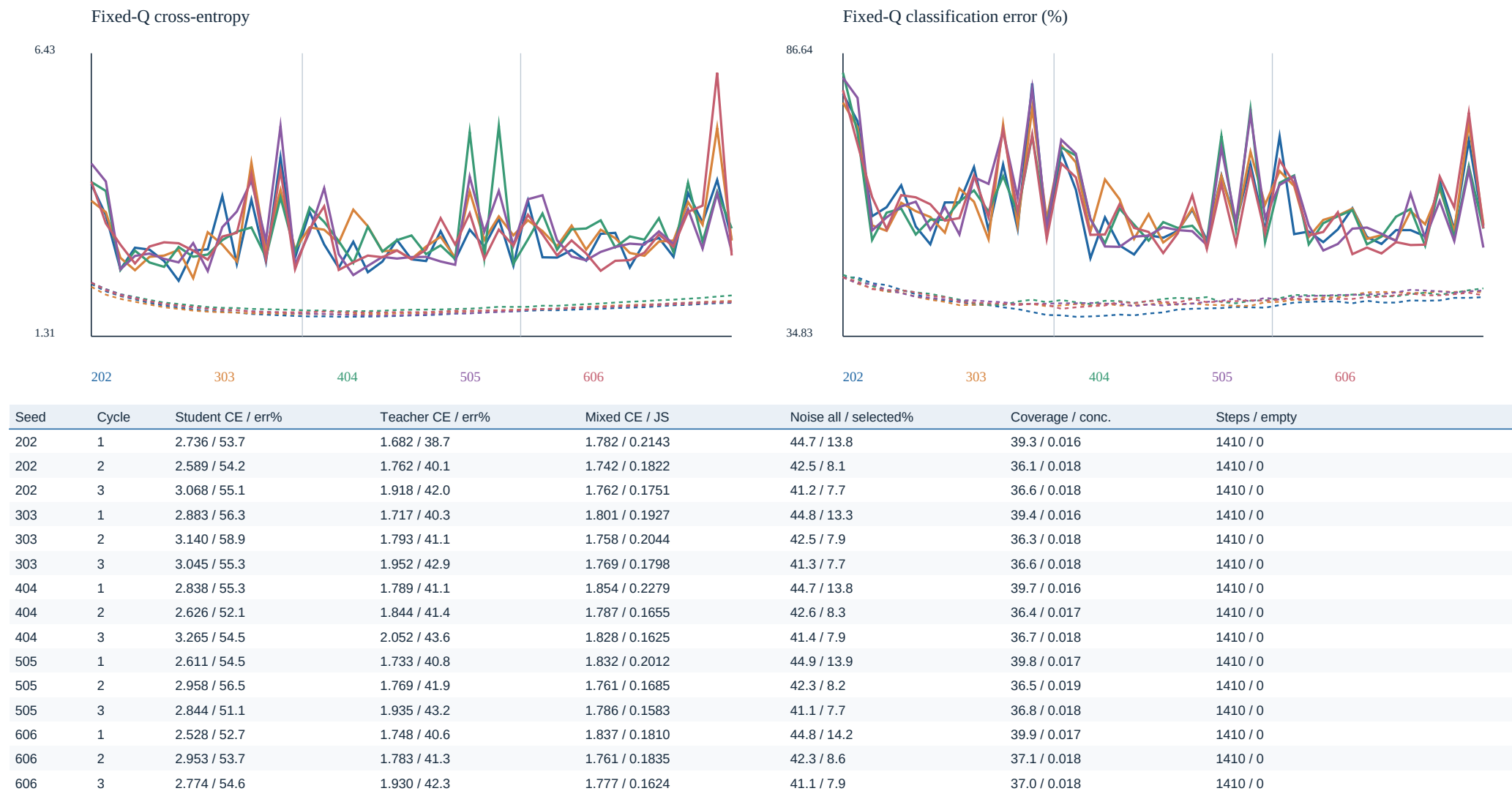
C4-1. Proxy versus labelled local update

Negative post-minus-pre CE means local improvement. Near zero means either absolute change is below 1e-4. Natural C4 executes updates without a separate accept/reject evaluator.

Seed	Probes	Both improve	Proxy down / local up	Proxy up / local down	Both worsen	Near zero
202	90	50	40	0	0	0
303	90	43	47	0	0	0
404	90	48	42	0	0	0
505	90	53	36	0	1	0
606	90	47	43	0	0	0

C4-2. Complete fixed-Q trajectories and pseudo-supervision

The fixed-Q panel uses the same 1,500 image-corruption evaluations at every block. Curves are unsmoothed. Dotted curves are teachers; solid curves are students. Missing/failed cells are absent, not assigned zero risk.



Cycle CSV retains teacher/student disagreement, mixed-target confidence, count-weighted pseudo-label statistics, selected-class concentration and elapsed block time. Selected noise is undefined when no examples were selected.

C4-3. Same-domain returns

Visit-3 minus visit-1 post-block student CE for each named corruption. Positive means worse on that domain; all domains and all terminal cells are retained. The companion block CSV includes pre/post CE and error for student and teacher at every visit.

Corruption	202	303	404	505	606
gaussian_noise	0.842	-0.302	-1.083	-1.067	-0.981
shot_noise	-0.608	0.176	0.306	-0.192	-0.245
impulse_noise	0.466	0.366	0.311	0.712	0.157
defocus_blur	0.326	0.423	0.139	0.309	0.279
glass_blur	-0.498	0.100	0.312	0.002	0.338
motion_blur	0.431	0.342	0.298	0.180	0.180
zoom_blur	0.273	0.140	0.232	0.202	0.032
snow	-0.589	0.408	0.534	0.091	-0.225
frost	-0.447	-0.717	0.382	0.701	0.437
fog	-0.322	0.196	0.486	0.179	-0.081
brightness	0.388	0.065	0.202	-0.069	0.260
contrast	-0.125	-0.650	0.193	-0.494	0.022
elastic_transform	0.629	0.451	0.044	0.039	0.662
pixelate	-0.764	1.328	-0.226	-0.709	0.203
jpeg_compression	-0.065	-0.028	0.274	0.348	0.169

C4-4. Per-seed parameter summary

Seed	Mean post CE / err%	Mean return dCE / dErr pp	Cycle-3 CE / err%	Coverage%	Updates	Block cost min
202	2.608 / 52.2	-0.004 / -4.71	2.637 / 50.2	37.3	4230	15.8
303	2.591 / 51.7	0.153 / -3.71	2.686 / 50.2	37.4	4230	15.1
404	2.654 / 51.9	0.160 / -3.10	2.650 / 49.9	37.6	4230	15.4
505	2.667 / 52.4	0.015 / -4.78	2.731 / 50.3	37.7	4230	15.0
606	2.623 / 51.8	0.080 / -2.65	2.708 / 51.1	38.0	4230	15.0

C4-5. Matched accept/reject and closed/replay forks

The first nonempty update after preselected blocks 5 and 15 is the focal candidate. Positive accept-minus-reject CE is worse. $A_{\text{closed}} = \text{accept_closed} - \text{reject_closed}$; $A_{\text{open}} = \text{accept_replay} - \text{reject_replay}$; $I_{\text{feedback}} = A_{\text{closed}} - A_{\text{open}}$. The H94 and H1410 forks share their initial state and schedule; their latest common scored checkpoint, H93, agrees exactly. H94 is therefore a correlated branch, not an independent replicate. The H1410 run does not separately score H94.

Seed	Parent	Immediate Q dCE / dErr pp	Immediate domain dCE / dErr pp
202	5	-0.021 / 0.40	-0.019 / -0.53
202	15	0.021 / -0.20	0.070 / 0.87
303	5	-0.077 / -0.40	-0.119 / -1.00
303	15	-0.049 / 0.07	0.068 / 1.00
404	5	-0.038 / -1.07	-0.045 / -1.07
404	15	0.014 / 0.47	-0.027 / -1.00
505	5	-0.051 / -0.07	-0.130 / -2.20
505	15	-0.049 / -0.40	-0.181 / -1.47
606	5	-0.035 / -0.27	-0.047 / -0.33
606	15	0.015 / -0.53	0.046 / 0.20

H94 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_{closed}	A_{open}	I_{feedback}
202	5	2.452	2.452	2.661	2.593	0.210	0.142	0.068
202	15	3.425	3.425	3.602	3.899	0.177	0.474	-0.297
303	5	2.768	2.768	2.776	2.775	0.008	0.006	0.001
303	15	3.852	3.852	3.220	3.259	-0.632	-0.592	-0.039
404	5	2.529	2.529	2.570	2.572	0.041	0.043	-0.002
404	15	3.340	3.340	3.591	3.939	0.251	0.598	-0.347
505	5	2.904	2.904	2.704	2.699	-0.201	-0.206	0.005
505	15	3.365	3.365	3.159	3.270	-0.206	-0.095	-0.111
606	5	3.072	3.072	3.068	2.999	-0.004	-0.073	0.069
606	15	3.422	3.422	3.222	2.968	-0.200	-0.455	0.254

H1409 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	2.638	2.638	2.472	2.687	-0.167	0.048	-0.215
202	15	2.595	2.595	2.589	2.978	-0.005	0.383	-0.389
303	5	3.211	3.211	3.304	3.089	0.093	-0.122	0.216
303	15	3.075	3.075	3.140	3.087	0.064	0.012	0.053
404	5	2.972	2.972	3.295	2.824	0.323	-0.149	0.471
404	15	3.000	3.000	2.626	2.907	-0.374	-0.093	-0.281
505	5	2.941	2.941	2.587	2.877	-0.354	-0.064	-0.290
505	15	3.023	3.023	2.958	3.264	-0.066	0.240	-0.306
606	5	2.791	2.791	2.773	2.953	-0.018	0.162	-0.179
606	15	2.607	2.607	2.953	2.639	0.346	0.032	0.314

H1410 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	2.578	2.578	2.479	2.702	-0.098	0.124	-0.222
202	15	2.633	2.633	2.628	3.029	-0.006	0.396	-0.402
303	5	3.311	3.311	3.595	3.272	0.283	-0.040	0.323
303	15	3.050	3.050	3.113	2.990	0.063	-0.060	0.123
404	5	2.973	2.973	3.250	2.780	0.277	-0.193	0.470
404	15	3.081	3.081	2.677	2.956	-0.405	-0.125	-0.279
505	5	2.882	2.882	2.561	2.938	-0.321	0.056	-0.376
505	15	3.023	3.023	3.022	3.382	-0.001	0.359	-0.360
606	5	2.812	2.812	2.791	2.894	-0.021	0.082	-0.103
606	15	2.644	2.644	3.008	2.653	0.364	0.010	0.355

C4-6. Diagnostic state, finite F and local identifiability

x =(student fixed-Q CE, mixed-target fixed-Q CE, student/teacher JS, mixed confidence). F is the mean H20 future x over three base tapes at each observed parent; it is a conditional response, not a closed transition law. The block CSV stores every observed x_t .

Seed	Parent	x_0	$F_{H20}(x_0)$	Tape SD	Finite	J status
202	5	2.880 / 2.021 / 0.277 / 0.696	2.586 / 2.008 / 0.239 / 0.695	0.042 / 0.000 / 0.008 / 0.000	132/132	not identified
202	15	2.736 / 1.782 / 0.214 / 0.677	3.301 / 1.780 / 0.303 / 0.676	0.336 / 0.000 / 0.043 / 0.000	132/132	not identified
303	5	2.751 / 2.011 / 0.253 / 0.697	2.873 / 1.999 / 0.264 / 0.696	0.151 / 0.000 / 0.017 / 0.000	132/132	not identified
303	15	2.883 / 1.801 / 0.193 / 0.677	3.856 / 1.800 / 0.312 / 0.676	0.515 / 0.000 / 0.021 / 0.000	132/132	not identified
404	5	2.649 / 2.076 / 0.255 / 0.699	2.546 / 2.065 / 0.223 / 0.698	0.104 / 0.000 / 0.004 / 0.000	132/132	not identified
404	15	2.838 / 1.854 / 0.228 / 0.675	3.464 / 1.851 / 0.317 / 0.675	0.046 / 0.000 / 0.011 / 0.000	132/132	not identified
505	5	2.810 / 2.063 / 0.250 / 0.701	2.702 / 2.052 / 0.246 / 0.700	0.099 / 0.000 / 0.008 / 0.000	132/132	not identified
505	15	2.611 / 1.832 / 0.201 / 0.676	3.490 / 1.830 / 0.280 / 0.676	0.216 / 0.000 / 0.012 / 0.000	132/132	not identified
606	5	2.938 / 2.063 / 0.259 / 0.700	2.638 / 2.052 / 0.229 / 0.700	0.057 / 0.000 / 0.015 / 0.000	132/132	not identified
606	15	2.528 / 1.837 / 0.181 / 0.678	3.245 / 1.835 / 0.280 / 0.678	0.028 / 0.000 / 0.023 / 0.000	132/132	not identified

Perturbation, conditioning and held-out diagnostics

Seed	Parent	Median G	Rank	Condition	Scale diff	Held-out median	Closure	Reason
202	5	13.3	4	3.3	0.74	21.12	falsified	not established
202	15	63.3	4	4.1	0.75	255.18	falsified	not established
303	5	34.2	4	3.2	0.83	96.82	falsified	not established
303	15	99.7	4	2.9	0.98	216.98	falsified	not established
404	5	22.7	4	3.6	1.60	111.60	falsified	not established
404	15	40.7	4	4.7	1.42	80.05	falsified	not established
505	5	25.9	4	3.2	0.96	80.13	falsified	not established
505	15	62.9	4	2.7	0.63	225.33	falsified	not established
606	5	19.6	4	4.4	0.88	77.93	falsified	not established
606	15	70.2	4	4.0	0.90	306.55	falsified	not established

Independent student/teacher perturbations, input rank/condition, two scales, held-out directions and momentum closure checks are retained in the compact analysis files. No eigenvalue or phase boundary is claimed without a closed and predictive state law.

C4-7. Bounded parameter-map reading

This checkpoint isolates one pair $\lambda=0.4$, $\beta=0.0005$ with five exploratory seeds. Seeds are reused across later pairs for paired comparison; they are not independent cross-parameter replications. Outcomes at this pair alone cannot locate a stability boundary. Numerical failures and improvement/null findings remain in the reported denominator.

Observed coverage: 5/5 complete 45-block trajectories, 10 verified H1410 parent nodes, 4 positive A_{closed} and 4 positive I_{feedback} . Parent nodes within one seed are correlated. A positive endpoint or interaction alone does not establish sustained recursive deterioration.

Coverage ledger and missingness

C4 item	Present evidence	Missing / limitation
C4-1	450 completed local probes	Unavailable after failed/incomplete trajectories
C4-2	5 full trajectories and cycle process records	Failed/partial cells not extrapolated
C4-3	75 same-domain return comparisons	Only complete first/third visits
C4-4	5 per-seed parameter summaries	One pair cannot characterize a sweep
C4-5	30 fork horizons	Unavailable parent nodes explicitly excluded
C4-6	10 H20 parent analyses	J/spectrum only if closure and held-out criteria pass
C4-7	One-pair status/sign description	No identified region or boundary

Provenance: Sapelo 48468709, compact evidence MANIFEST.json and AUDIT.json. Raw checkpoints, logits, step tapes and all failures remain under /scratch/yz54720/RSI2RSD/C4/sweep_yingchuan_20260923/results/I04_b00005/. The reserved confirmation panel was not read.